

Homework 3

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Instructions: Include all codes and R-outputs that are necessary for each question. Submission: You MUST submit a PDF file.

Question 1: Regression Models: Is there a relationship between student math test scores and socioeconomic variables? The data set `CASchools.csv` contains data on test performance, school demographics, and student demographic background for school districts in California. Remove the variable `county` before the analysis.

Please use the Data set description document to learn more about the data set.

a). Fit a linear regression model to predict math test scores using the variables in the dataset. Discuss your results, making sure to cover the following points:

1. What do you observe about the relationship between these predictors and math test scores?
2. Are there any insignificant variables?
3. Which predictor is explaining math scores the best?
4. Explain the interpretation of the coefficient on lunch in terms of the response variable.

b). Do you think you can fit a better model than the model in part (a)? Fit a new model that you think is a better fit for the data. What do you observe about this new model? (Use an appropriate variable selection algorithm and use cross-validation to validate your answers).

c). Write the equation for the least square fit from part b).

d). Compare the R² and Testing MSE for models from part (a) and (b).

e). Using a residuals plot for part a and part b models, state the appropriateness of using a linear model for this problem.

f). Incorporate an interaction term into your multiple linear regression and respond to the following:

1. Provide analytical justification for your interaction term.
2. Compare your results with the models from parts a) and b).

g). Fit a 'Ridge Regression' Model for the dataset and interpret the results.

h). Fit a 'LASSO Regression' model for the dataset and interpret the results.

i). Compare all Testing MSE values that you got from parts a,b,f,g and h. Which model do you recommend?

Problem 2: Tree based models: This question should be answered using the “*Wine Quality*” data set. Description about the data set can be found on the link provided. Objective of this question is to fit an regression tree model to predict quality of wine.

a). Produce some numerical and graphical summaries of the data set. Explain the relationships.

b). Create a training set with 80% of the observations, and a testing set containing the remaining 20%.

- c). Fit a regression tree with *quality* as the response variable using the training set. Plot the tree and interpret the results. What test MSE do you obtain?
- d). Use cross-validation in order to determine the optimal level of tree complexity. Does pruning the tree improve the test MSE?
- e). Use random forests to analyze this data. What test MSE do you obtain?
- f). Use the *importance()* function to determine which variables are most important.